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# A Review of Nigerian Bentonitic Clays as Drilling Mud

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## Abstract

The heydays of bentonitic clay production in Nigeria occurred in the late 1950s when oil was first discovered. This was short lived as the advent of imported bentonite took the centre stage in the early 1960s. Since then, there has been no widespread use of the Nigerian bentonitic clay as drilling muds in Nigeria. In the recent past, precisely in 2001, the only major use of Nigerian bentonite for drilling oil wells was made by Shell Petroleum Development Company, Nigeria where over 600 wells were drilled with it. In recent years, Nigerian researchers have been actively pursuing the study of using Nigerian clay as drilling muds and finding potential markets for it. Many of these studies have focused on characterizing and beneficiating the clays to the American Petroleum Institute (API) standard. The available studies in this direction have been chronicled in this paper with the major research findings of each individual researcher brought to the fore. Much of the information that has been gathered in this review indicate that all the researchers were in unison in their submission that the Nigerian bentonitic clay is predominantly Calcium based and that it requires some measure of beneficiation to be effective for use as drilling muds. Mention is also made of possible gray areas where more research efforts should be directed. This work is predicated upon the belief that the works of the past combined with that of the present can lay the necessary framework for future studies on Nigerian bentonitic clays. The review this paper presents can serve as a baseline for the oil and gas E & P companies desirous of unlocking the potential of using Nigerian bentonitic clays as drilling muds.

**Keywords : Bentonite, Nigerian clays, Beneficiation, Drilling mud**

## Introduction

As the world leans on oil and gas as the major source of energy, industry operators, in a bid to satisfy this demand have become positively restless. Today, as a result of their restlessness, oil and gas exploration and production activities dot virtually every part of the globe where hydrocarbon deposits are deemed to occur. In order to produce these hydrocarbons, the industry relies on a number of oilfield chemical formulations. One of such chemical formulations is the drilling fluid. In recent times, the demand for this fluid has gone sky high. Little wonder, in 2011, *Freedonia*, a market research organization reported that of all oilfield chemicals, the drilling fluid was and will still be of high demand. As at then it gulped about 42% of the total cost of oilfield chemical expenditure of $18.2 billion as shown in Figure 1 (Caenn, 2013).

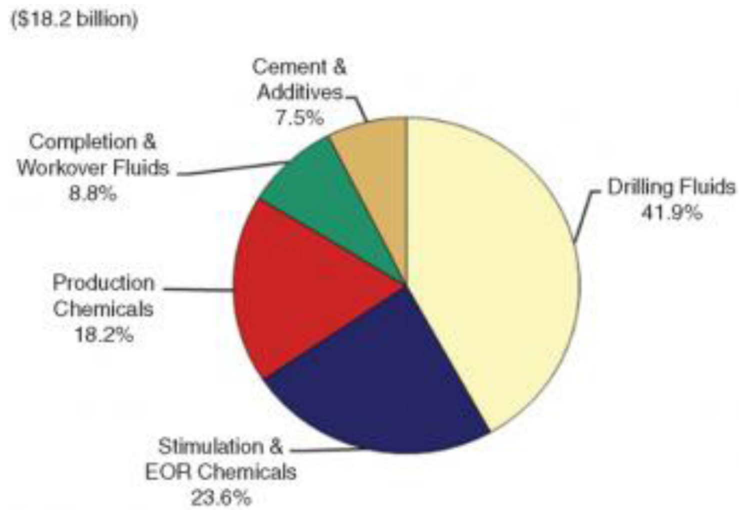


**Figure 1—World Oil Field Chemical Demand (Freedonia Group in Caenn, 2013)**

In addition to the aforementioned, a research report from *Markets and Markets.com* projected revenues from drilling and completion fluids through from 2011 through 2018 as shown in Figure 2. As can be seen from the "stair-step" effect of the chart, total revenues were projected to increase each year from 2011 to 2018 (Caenn, 2013).

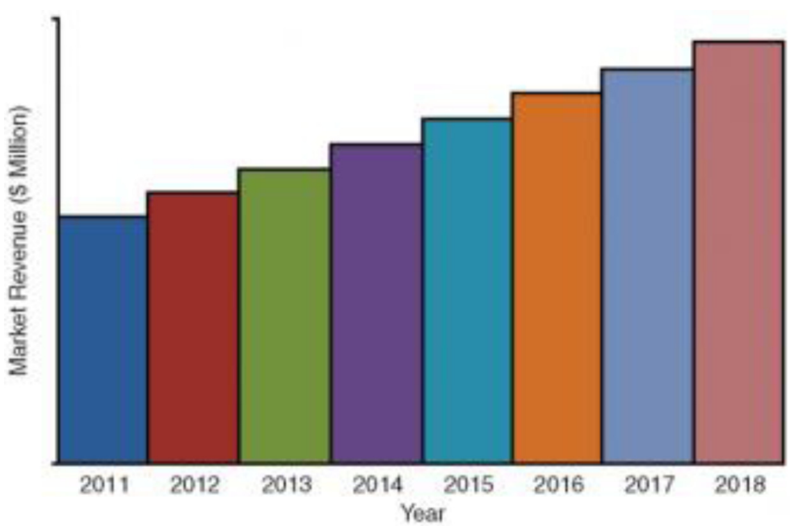


**Figure 2—Drilling and Completion Fluids Market by Revenue from 2011-2018 [marketsandmarkets Analysis in Caenn, 2013]**

As earlier alluded to, the increasing revenues from drilling and completion fluids as predicted by *markets and markets* may partly be attributed to the increasing number of discoveries of oil and gas reserves worldwide. This in turn increases drilling activities which require the use of drilling fluids. Again, if the research findings on the market demand for drilling fluid by *Freedonia* are anything to go by, then it would be safe to say that as drilling fluid demand increases, the demand for bentonite clay (one of the principal components of most drilling fluids) would correspondingly get high. With this anticipated increase in demand, pressure would pile on the existing bentonitic clay reserve sources; hence depleting it at a rate faster than hitherto anticipated. On the local scene, Odumugbo (2005) asserts that the present consumption of bentonite clay in the drilling operations in Nigeria is put at over 50 thousand tons a year and all of it is imported from USA. It must, however, be noted that bentonite when added to fresh water to make a drilling fluid plays a number of invaluable roles namely: to increase the hole cleaning properties of the fluid, reduce water seepage or filtration into permeable formation, form a thin filter cake of low permeability, promote hole stability in poorly cemented formations and avoid or overcome loss of circulation (Chilingarian and Vorabutr, 1983; Elward-Berry and Darby, 1997). Due to the invaluable nature of this clay and the anticipated supply shortages emanating from the rising demand, many laboratory studies have been directed towards the search for local bentonitic clay deposits which can be enhanced to the American Petroleum Institute (API) standard for bentonite. This search has yielded fruits as several parts of the globe have presented laboratory studies of clays from their regions that have the potential of being used as an alternative to the API bentonite. One area of the globe where there is an avalanche of “untiring” researchers who have given the subject of finding and standardizing clays in their area to the API specification for drilling muds is Nigeria. Nearly all areas of the Nigerian state have had their clays studied in whole or in part for use as an alternative to the API bentonite as indicated in Figure 3. Previous researches in this direction indicate that a bentonite deposit of a probable reserve of about 700 million tonnes has been found in North-eastern Nigeria even though the clay in its natural form has properties below the standard required for drilling fluid application (RMRDC, 1990; RMRDC, 1991; Aribisala, 1993). As reported by Terra mines in Edo state, Nigeria, the bentonite deposits found in the Niger Delta region of Nigeria holds a probable reserve of over 4 billion tonnes (Emofurieta, 2001). Over the last three decades, many scholarly papers have been published in both local and international journals that address this subject. Most of these publications give qualitative studies on what has been done in enhancing the local clays to meet the standards of the API bentonite. However, due to the elaborate nature of the studies that have been presented by diverse researchers, there is need to assess all the studies and the research findings in one piece so as to isolate what has been achieved and identify possible areas of interest where more studies should be directed. In this paper therefore, a chronicle is made of the available studies on the characterization and enhancement of Nigerian clays for use as drilling muds, the various research findings as presented by diverse authors and the future research needs on the Nigerian bentonitic clay is presented.

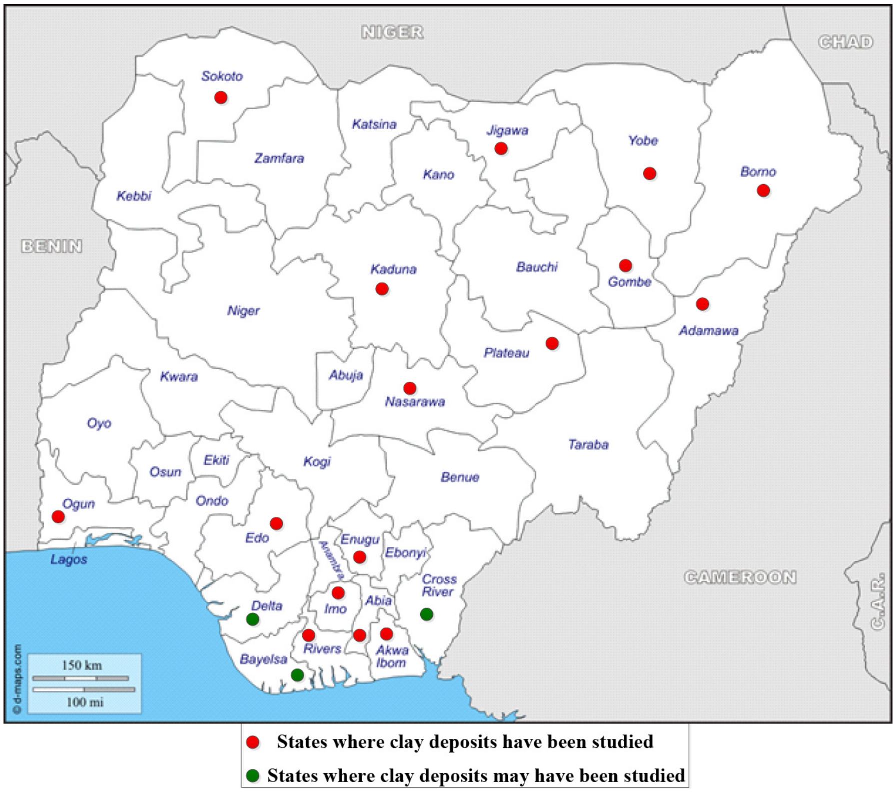


**Figure 3—Map of Nigeria showing states where bentonitic clay has been studied [Photo credit: d-maps.com]**

## Studies on Nigerian Clay for use as Drilling Muds

Studies on Nigerian clays for use as drilling muds would be categorized in this paper as: The pioneering years, the years of slow activity and the awakening years.

**The pioneering years [1959 – 1962]**

From available literature, studies on Nigerian clay for use as drilling mud dates back to the early days when oil was first discovered in Nigeria. Polo (1962) reported that bentonitic clays, which are used for drilling mud, were first mined and pulverized at Isieke Ihekie in Bende Local Government Area of Abia state in 1961. The capacity then was 3000 bags per day. He further added that 5016, 4000 and 1700 metric tons of drilling muds were produced from Nigerian clays in the years 1961, 1962 and 1963 respectively. Further studies on Nigerian clays indicate that other parts of the country possess clay deposits of montmorillonite content. This was shown by Oyawoye and Hirst (1964) whose works indicate that a montmorillonite mineral was present at Ropp, Plateau province in Northern Nigeria. The work of Porrenga also showed that clay from the Niger Delta had montmorillonite content which increases with water depth from the shore.

**The years of slow activity [1960 – Late 90's]**

The period between 1960 and 1990 was a period of slow activity in this research area. This is evident by the few scholarly works by researchers during this period. The low research efforts during these times and the rising mud quantity requirements of the oil industry led to the importation of bentonite from the United States. Bindei (1987) asserted that drilling operations in Nigeria began in the mid-fifties and local additives and clays were used on drilling fluids. Later in the early 1960s, the use of local additives and clays for drilling in the petroleum industry subsided in Nigeria as a result of the introduction of imported commercial additives and bentonite. Again, the effect of the three yearlong Nigeria – Biafra civil war between 1967 and 1970 may have also impeded research motion during this time. After this period, some researches resurfaced again. Omole *et al.* (1989) reported their research findings in the area of beneficiating and using local clays as drilling muds. Further work in finding bentonitic clay reserves came alive again during the early 1990's. During this period, it is seen that the works that were mostly available were those reported by the Raw Materials Research and Development Council (RMRDC), Abuja. Their research was mainly targeted towards finding bentonitic reserve locations around the country. Their drive in this search was energized by the fact that the reliance on importation was causing a major drain on the Nigerian economy leading to local mineral resource under-utilization and capital flight.

**The awakening years [2000 - Present]**

With the reserve data by the researchers in the period of slow activity, researchers launched a massive hunt for these clays. By the early 2000's an avalanche of published literature on studies of Nigerian clay for use as drilling muds surfaced. This upsurge in published research on the subject caused a major awakening. This awakening led Shell Petroleum Development Company (SPDC) Nigeria to drill over 600 wells with the Nigerian clay in 2001(Emofurieta, 2001). Capitalizing on this development and drawing strength from other individual researchers and their earlier efforts in the early 90's, the Raw Materials Research Development Council (RMRDC), Abuja in 2010 assessed the location of bentonite deposits in Nigeria and their estimated reserve quantities. Their findings are presented in Table 1.

**Table 1—Location of bentonite deposits in Nigeria [RMRDC, 2010]**

| S/N | STATE | LOCATION | ESTIMATED RESERVE | REMARKS |
|---|---|---|---|---|
| 1. | Cross River | Ogurude | - | Needs further investigation |
| 2. | Akwa Ibom | Itu | - | Needs further investigation |
| 3. | Ebonyi | Ohaozara | - | Products of Asu River group shale |
| 4. | Abia | Arochukwu, Umuahia, Bende, Isiukwuato, Ikwuano | Proven /inferred reserve estimates is 5.8 – 7.5 million tonnes | Most deposits require further investigation |
| 5. | Imo | Orlu, Isu, Oru, Okigwe | Proven /inferred reserve estimates is 5.8 – 7.5 million tonnes | Mining activities are on in some of the locations |
| 6. | Anambra | Awka | Not yet quantified | Needs further investigation |
| 7. | Gombe | Akko, Gombe, Yamaltu-Deba | - | - |
| 8. | Adamawa/Taraba | Gujba(Mutai) | Not yet quantified | Deposits in River channel |
| 9. | Yobe | Ngala, Marte, Mongunu, Damboa | - | Black cotton |
| 10. | Borno | Gamboru, Marte, Ngala, Dikwa, Mongunu | 700 million tonnes | Product of clay of quater-ry and tertiary |
| 11. | Edo | Akoko-Edo, Afuze, Okpebhio, Esan, Owan, Etsako | - | - |
| 12. | Kebbi | Jega | Not yet quantified | Found in Benin formation |

By the late 2000s, so much work had been done by the researchers especially in studying the clays mainly through characterization and/or beneficiation. This dominated the rest of the period up until the present time. Thus, the works of these researchers from the pioneering years up until the present time are summarized and presented in Tables 2 through 6. Their area of study, the experiments they did, the results they obtained as well as their major findings are documented and discussed in detail in the following section.

**Table 2—Studies by Researchers on Nigerian Bentonitic Clays**

| Year | Researcher(s) | Area studied | Number of locations studied | Specific areas studied | Experiments done | Materials used for upgrade |
|---|---|---|---|---|---|---|
| 1964 | Oyawoye & Hirst | Northern Nigeria | 1 | Ropp, plateau province | XRD, thermal analysis | - |
| 19xx | Porrenga, D.H. | Niger Delta | - | - | XRD, XRF | - |
| 1989 | Omole *et al.* | Borno State | 20 | Bama-Mubi, Maiduguri-Bama, Maiduguri-Gamboru, Dikwa-Marte, Dikwa-Maiduguri, Maiduguri-Biu, Biu-Numan, Yola-Fufore, Ngurore, Numan-Gombe, Numan Roads | XRD, Rheological & filtration, physico chemical properties | $Na_2CO_3$, Trona |
| 1999 | Ademibawa, A.T. | Adamawa State | 1 | Pindiga | XRD | - |
| 2003 | Onize, D.E. | Edo & Borno States | 2 | Afuze & Maiduguri | Rheological analysis | CMC, PAC, Soda ash |
| 2005 | Odumugbo, C.A. | Edo state | 1 | Afuze | Rheological analysis | Starch, Potash |
| 2006 | Okorie, O.M. | - | - | - | Rheological analysis | Trona, Starch, BPSHP, Soda ash, Lime, NaOH |
| 2007 | Falode *et al.* | Gombe State | 1 | Pindiga | Rheological & filtration tests | $Na_2CO_3$, Potash, Starch |
| 2008 | James *et al.* | Adamawa State | 1 | Yola | XRF, XRD, Rheological analysis | $Na_2CO_3$ |
| 2010 | Salam *et al.* | Ogun State | 1 | Ewekoro | Rheological analysis, Model development | Local Gum Arabic, $Na_2CO_3$ |
| 2010 | Lafia, I.U. | Nasarawa State | 1 | - | Rheological analysis | Starch |
| 2010 | Joel & Nwokoye | - | - | - | Rheological analysis, Chemical & Economic analysis | - |
| 2011 | Abdullahi *et.al* | Benue trough | 1 | Fika | XRF, XRD, Rheological analysis | - |
| 2011 | Dewu *et.al* | Gombe state | 3 | Pindiga<br>Futuk<br>Arawa River | XRF, XRD | Soda Ash + viscosifier |
| 2011 | Osadebe *et.al* | Edo State | 1 | Okada | XRD | None |
| 2011 | Apugo-Nwosu *et al.* | Abia State | 1 | Ubakala | XRD, XRF | $Na_2CO_3$, PAC, CMC |
| 2012 | Onwuachi-Iheagwara, P.N. | Imo & Delta | 3 | Akokwa<br>Akpehe-Olomu<br>X | Rheological analysis | - |
| 2012 | Dewu *et al.* | Yobe State | 3 | Ngalda<br>Maiduwa 1<br>Maiduwa 2 | XRF, XRD, AAS | $Na_2CO_3$, Drispac Polymer |
| 2012 | Ahmed *et al.* | Gombe State | 1 | Pindiga | XRD, XRF | Conc. $H_2SO_4$ |

**Table 2—Continued.**

| Year | Researcher(s) | Area studied | Number of locations studied | Specific areas studied | Experiments done | Materials used for upgrade |
|---|---|---|---|---|---|---|
| 2012 | Udoh & Okon | Akwa Ibom State | 1 | Ewet Offot, Uyo | Rheological analysis | $Na_2CO_3$, Sweet potato starch |
| 2013 | Omole *et al.* | Kaduna & Sokoto States | 10 | 12° 56' 51'' N 005° 12' 01'' E | Rheological and Filtration experiments | $Na_2CO_3$ |
| | | | | 12° 29' 30'' N 007° 59' 13'' E | | |
| | | | | 12° 55' 48'' N 005° 18' 11'' E | | |
| | | | | 12° 52' 36'' N 005° 15' 56'' E | | |
| | | | | 12° 52' 19'' N 005° 20' 10'' E | | |
| | | | | 12° 50' 36'' N 005° 21' 04'' E | | |
| | | | | 13° 06' 44'' N 005° 16' 32'' E | | |
| | | | | 13° 44' 42'' N 005° 39' 24'' E | | |
| | | | | 13° 45' 31'' N 005° 36' 48'' E | | |
| | | | | 13° 45' 11'' N 005° 38' 49'' E | | |
| 2013 | Obaje, S.O. | Borno State | 2 | New Marte (NM) | AAS | - |
| | | | | Dikwa (D4) | | |
| 2013 | Olugbenga *et al.* | Niger Delta | 1 | - | XRF, XRD | $Na_2CO_3$ |
| 2013 | Nwosu *et al.* | Enugu State | 1 | Udi | XRD, FTIR | - |
| 2013 | Oyegoke, T. | Adamawa State | 1 | Sangere, Yola | Rheological analysis, Model development | Polypac PAC R, Soda ash |
| 2014 | Imuentinyan & Adewole | Edo State | 2 | Ovia | Rheological analysis | CMC, Tannin |
| | | | | Ikpoba | | |
| 2014 | Nmegbu, C.G.J. | Rivers State | 3 | Egbamini (EMOLGA) | Rheological analysis | NaOH, Quick troll, CMC, $Na_2CO_3$ |
| | | | | Afam Street (PHALGA) | | |
| | | | | Oboboru (ONELGA) | | |

**Table 3—Studies on Nigerian Clays Showing Elemental Analysis Results**

| | | | Percentage of Elements | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | **Researcher(s)** | **Area Studied** | **$SiO_2$** | **$Al_2O_3$** | **$Fe_2O_3$** | **MgO** | **CaO** | **$K_2O$** | **$Na_2O$** | **$TiO_2$** | **$P_2O_5$** | **MnO** | **LOI** |
| 1964 | Oyawoye & Hirst | Ropp, Plateau province | 47.38 | 21.27 | 10.66 | 0.42 | 0.78 | 0.08 | 0.12 | - | - | - | 9.6 |
| 19xx | Porrenga, D.H. | Niger Delta | 43.5 | 24 | 10.6 | 2 | 0.8 | 1.5 | 0.3 | 0.7 | - | - | 1.6 |
| 1999 | Ademibawa, A.T. | Pindiga | 53.06 | 12.09 | 2.71 | 1.10 | 1.30 | 1.25 | 6.4 | 1.53 | - | 0.08 | 20.19 |
| 2008 | James *et al.* | Yola | 61.35 | 14.09 | 7.14 | 0.87 | 0.12 | 1.05 | 1.15 | 1.453 | 0.06 | 0.023 | 12.23 |
| 2010 | Joel & Nwokoye | - | - | - | 0.0096 | 0.014 | 0.24 | - | 0.27 | - | - | - | - |
| 2011 | Abdullahi *et al.* | Fika | 35.17 | 13.05 | 7.35 | 6.17 | 10.36 | 1.69 | 0.06 | 0.64 | 0.31 | 0.27 | 15.73 |
| 2011 | Dewu *et al.* | Pindiga | 53 | 18 | 4.68 | 2.6 | 0.67 | 2.28 | 0.61 | 0.9 | - | - | 15.12 |
| | | Futuk | 46.76 | 18.4 | 8.17 | 2.34 | 0.57 | 2.59 | 0.45 | 0.7 | - | - | 14.82 |
| | | Arawa River | 52.8 | 20.8 | 7.53 | 2.95 | 0.77 | 2.61 | 0.28 | 0.67 | - | - | 15.63 |
| 2011 | Osadebe *et al.* | Okada | 55.77 | 20.6 | 0.70 | 0.2 | 0.3 | 0.3 | 2 | 1.15 | 0.012 | 0.02 | 16.8 |
| 2011 | Apugo-Nwosu, *et al.* | Ubakala | 69.6 | 16 | 2.99 | 0.156 | 0.22 | 0.599 | 0.056 | 2.64 | - | - | - |
| 2012 | Ahmed *et al.* | Pindiga | 43.6<br>46.9* | 14<br>15* | 26.54<br>24.15* | - | 2.46<br>2* | 3.52<br>3.71*<br>($K_2O$ +-$_2$O + MgO) | | 2.06<br>2.15* | - | 1 | 6.46<br>6.63<br>* |
| 2012 | Dewu *et al.* | Ngalda | - | *7.32* | - | *2.71* | *2.68* | - | *1.25* | *0.45* | - | - | *9.73* |
| | | Maiduwa 1 | - | *4.64* | - | *3.56* | *5.59* | - | *2.43* | *0.28* | - | - | *8.77* |
| | | Maiduwa 2 | - | *6.51* | - | *4.19* | *5.07* | - | *2.04* | *0.28* | - | - | *9.24* |
| 2013 | Omole *et al.* | 12° 56' 51'' N<br>005° 12' 01'' E | - | - | - | 4.68 | 36.38 | 0.25 | 0.65 | - | - | - | - |
| | | 12° 29' 30'' N<br>007° 59' 13'' E | - | - | - | 3.01 | 49 | 0.05 | 1.8 | - | - | - | - |
| | | 12° 55' 48'' N<br>005° 18' 11'' E | - | - | - | 6.01 | 26.73 | 0.19 | 0.82 | - | - | - | - |
| | | 12° 52' 36'' N<br>005° 15' 56'' E | - | - | - | 6.25 | 41.35 | 0.27 | 0.43 | - | - | - | - |
| | | 12° 52' 19'' N<br>005° 20' 10'' E | - | - | - | 2.12 | 42.05 | 0.21 | 0.49 | - | - | - | - |
| | | 12° 50' 36'' N<br>005° 21' 04'' E | - | - | - | 4.67 | 43.75 | 0.18 | 1.07 | - | - | - | - |
| | | 13° 06' 44'' N<br>005° 16' 32'' E | - | - | - | 5.3 | 37 | 0.13 | 0.71 | - | - | - | - |
| | | 13° 44' 42'' N<br>005° 39' 24'' E | - | - | - | 26.4 | 27.95 | 0.46 | 1.15 | - | - | - | - |
| | | 13° 45' 31'' N<br>005° 36' 48'' E | - | - | - | 37.69 | 18.70 | 0.15 | 1.11 | - | - | - | - |
| | | 13° 45' 11'' N<br>005° 38' 49'' E | - | - | - | 5.5 | 30.65 | 0.13 | 0.72 | - | - | - | - |
| 2013 | Obage, S.O. | New Marte (NM) | 24.21 | 60.05 | 0.03 | 3.02 | 0.6 | 1.01 | 1.98 | - | - | Trace | 9.1 |
| | | Dikwa (D4) | 26.30 | 58.40 | 0.01 | 3.22 | 0.45 | 0.98 | 2.01 | - | - | Trace | 7.71 |
| 2013 | Olugbenga *et al.* | Niger Delta | 47.4 | 20.97 | 5.73 | 9.48 | 4.23 | 0.98 | - | 1.18 | - | 0.01 | - |
| 2013 | Nwosu *et al.* | Udi | 53.2 | 5.906 | 0.444 | - | 0.006 | 0.121 | 1.417 | - | - | - | 4 |

**Table 4—Studies on Nigerian Clays Showing Rheological Properties of Beneficiated Clays**

| Year | Researcher(s) | Area Studied | PV | AV | YP | Fluid loss | pH | Density | Gel strength |
|---|---|---|---|---|---|---|---|---|---|
| 1989 | Omole *et al.* | North Eastern Nigeria | 0.44-1.57 | 3.3-7.45 | 2.69-6.33 | 26.5-63 | 6.1-8.8 | - | - |
| 2003 | Onize, D.E. | Afuze | - | 37 | - | 8.5 | 10 | - | - |
| | | Maiduguri | - | 50 | - | 9.5 | 10 | - | - |
| 2007 | Falode *et al.* | Pindiga | 2 -18 | - | - | 10 | - | - | - |
| 2008 | James *et al.* | Yola | - | 1.402 | 2.74 | - | 10.2 | - | - |
| 2010 | Salam *et al.* | Ewekoro | 1.25 – 28.83 | 1.5 - 52 | 1-7.83 | - | - | - | - |
| 2010 | Joel & Nwokoye | - | 7 - 11 | - | 4 - 15 | - | - | - | - |
| 2011 | Abdullahi *et al.* | Fika | - | - | 1-6 | <15 | - | - | - |
| 2011 | Dewu *et al.* | Pindiga | 7 - 22 | - | - | - | - | - | - |
| | | Futuk | 5.7 - 21 | - | - | - | - | - | - |
| | | Arawa River | - | - | - | - | - | - | - |
| 2011 | Osadebe *et al.* | Okada | *55.77* | *20.6* | *0.70* | *0.2* | *0.3* | *0.012* | *0.02* |
| 2011 | Apugo-Nwosu, *et al.* | Ubakala | 3 - 20 | - | 3-15 | - | - | - | 4/18 |
| 2012 | Onwuachi-Iheagwara, P.N. | Akokwa | *8.3* | *10.52* | *4.43* | - | *9* | *8.71* | *0.37/6.13* |
| | | Akperhe-Olomu | *13.9* | *16.75* | *5.7* | - | *12* | *8.6* | *0.37/0.47* |
| | | X | *14.3* | *16.48* | *4.3* | - | *10* | *8.6* | *0.37/4.26* |
| 2012 | Dewu *et al.* | Ngalda | - | - | 1.9 | - | 14 | - | - |
| | | Maiduwa 1 | - | - | 1.8 | - | 14 | - | - |
| | | Maiduwa 2 | - | - | 2 | - | 14 | - | - |
| 2012 | Udoh & Okon | Ewet offot, Uyo | 7 | 11.5 | 9 | - | 9.6 | 9.2 | 3/5 |
| 2013 | Omole *et al.* | 12° 56' 51'' N 005° 12' 01'' E | - | - | - | 51.6 | 9.61 | 8.76 | 1/1.4 |
| | | 12° 29' 30'' N 007° 59' 13'' E | - | - | - | 50.3 | 10.29 | 8.83 | 1.4/2 |
| | | 12° 55' 48'' N 005° 18' 11'' E | - | - | - | 54.2 | 10.05 | 8.77 | 0.75/0.88 |
| | | 12° 52' 36'' N 005° 15' 56'' E | - | - | - | 49.75 | 9.49 | 8.76 | 1.25/1.75 |
| | | 12° 52' 19'' N 005° 20' 10'' E | - | - | - | 42.25 | 9.74 | 8.83 | 1.875/2.75 |
| | | 12° 50' 36'' N 005° 21' 04'' E | - | - | - | 48.70 | 10.05 | 8.76 | 0.75/1.25 |
| | | 13° 06' 44'' N 005° 16' 32'' E | - | - | - | 44.05 | 9.89 | 8.76 | 1.5/2.25 |
| | | 13° 44' 42'' N 005° 39' 24'' E | - | - | - | 45.65 | 9.51 | 8.78 | 1.125/1.5 |
| | | 13° 45' 31'' N 005° 36' 48'' E | - | - | - | 41.58 | 10.16 | 8.79 | 2.75/3.625 |
| | | 13° 45' 11'' N 005° 38' 49'' E | - | - | - | 49.58 | 9.67 | 8.79 | 1.25/1.4 |
| 2013 | Oyegoke, T. | Sangere, Yola | 17.3 | 26.59 | 18.56 | 13.35 | 9.5 | - | - |
| 2013 | Nwosu *et al.* | Udi | - | - | - | 49.58 | 5.63 | 1.67 | - |
| 2014 | Imuentinyan & Adewole | Ovia | 10 | - | 7 | 11.4 – 16.5 | 8 | 8.6 | - |
| | | Ikpoba | 12 | - | 10 | 18.8 – 19.6 | 8 | 8.4 | - |
| 2014 | Nmegbu, C.G.J. | Egbamini; Afam street; Oboboru | 1.06-15.55 | 1.42-22.61 | 0.88-15.22 | 14.6-39.27 | 10 | 8.28-8.47 | 1.99-56.55 |

**NB: Values in italics are average values**

**Table 5—Studies on Nigerian clays showing major findings by researchers**

| Year | Researcher(s) | Area studied | Specific areas studied | Major Findings |
|---|---|---|---|---|
| 1964 | Oyawoye & Hirst | Plateau State | Ropp, Plateau province | Montmorillonite mineral belonging to the beidellite-nontronite series occurs due to high Al and Fe content |
| 19xx | Porrenga, D.H. | Niger Delta | - | Kaolinite, montmorillonite & small amount of illite. Montmorillonite content increases with water depth & distance to the shore |
| 1989 | Omole *et.al.* | Borno State | See Table 2 | Major clay mineral across the locations studied is Smectite. The clays are not suitable for oil drilling muds but with beneficiation with Trona and Soda ash, it can be used for drilling water wells. |
| 2003 | Onize, D.E. | Edo & Borno States | Afuze & Maiduguri | Beneficiated clays met API requirements for drilling mud |
| 2007 | Falode *et.al.* | Gombe State | Pindiga | Beneficiated clay met API specifications. A combination of potash & starch is good candidate for improvement of the rheological properties of the clays |
| 2008 | James *et.al.* | Adamawa | Yola | Calcuim based bentonite which can be activated to API mud standard. |
| 2010 | Salam *et.al.* | Ogun State | Ewekoro | Simultaneous increase in concentration of sodium carbonate and Gum Arabic led to a tremendous increase in apparent viscosity of clays |
| | | | | |
| 2010 | Joel & Nwokoye | - | - | Calcium based montmorillonite |
| 2011 | Abdullahi *et.al.* | Benue trough | Fika | Montmorillonite, Kaolinite, Dolomite, Ankerite, Quartz and Microcline |
| 2011 | Dewu *et.al.* | North Eastern Nigeria (Gombe state) | Pindiga | Calcium based montmorillonite |
| | | | Futuk | Montmorillonite, Kaolinite, Dolomite, illite, quartz, microcline |
| | | | Arawa River | Montmorillonite, Kaolinite, Dolomite, illite, quartz, microcline |
| 2011 | Osadebe *et al.* | Edo State | Okada | Montmorillonite, quartz, tin, roaldite, rankinite, kaolinite, antigorite, tantalum, pyrolusite, graphite, lime |
| 2011 | Apugo-Nwosu, *et al.* | Abia State | Ubakala | Montmorillonite, kaolinite quartz, Biotite, Calcite Feldspar |
| 2012 | Dewu *et al.* | Yobe State | Ngalda<br>Maiduwa 1<br>Maiduwa 2 | The samples collected were found to be Ca-based which require some beneficiation. After beneficiating the samples using $Na_2CO_3$, the rheological properties of the samples improved considerably and when a polyanionic polymer, Trade Name Drispac, was added, the rheological properties improved dramatically to a level that is typical of the standard bentonite. |
| 2012 | Onwuachi-Iheagwara, P.N. | Imo & Delta State | Akokwa<br>Akpehrhe-Olomu<br>X | Clays from the studied sites are suitable for use as drilling muds |
| 2012 | Ahmed *et al.* | Gombe State | Pindiga | The Pindiga raw bentonite was found to contain Ca-, K-, Na- and Mg-montmorillonite minerals with the calcium type being the predominant mineral. The calcium type also appeared to be the most thermally stable |
| 2012 | Udoh & Okon | Akwa Ibom state | Ewet Offot, Uyo | The Ewet-Offot clay when beneficiated at a considerable concentration with soda ash & sweet potato starch gives a promising potential as drilling mud clay |

Table 5—Continued

| | | | | |
|---|---|---|---|---|
| 2013 | Omole *et al.* | Kaduna & Sokoto states | 12° 56' 51'' N 005° 12' 01'' E | Calcium – Smectite |
| | | | 12° 29' 30'' N 007° 59' 13'' E | Calcium - Smectite |
| | | | 12° 55' 48'' N 005° 18' 11'' E | Illite, chlorite, kaolinite |
| | | | 12° 52' 36'' N 005° 15' 56'' E | Calcium - Smectite |
| | | | 12° 52' 19'' N 005° 20' 10'' E | Calcium - Smectite |
| | | | 12° 50' 36'' N 005° 21' 04'' E | Calcium - Smectite |
| | | | 13° 06' 44'' N 005° 16' 32'' E | Calcium - Smectite |
| | | | 13° 44' 42'' N 005° 39' 24'' E | Mixed |
| | | | 13° 45' 31'' N 005° 36' 48'' E | Vermiculite |
| | | | 13° 45' 11'' N 005° 38' 49'' E | Illite, chlorite, kaolinite |
| 2013 | Obage, S.O. | Borno State | New Marte (NM) | Clays studied have high montmorillonic content suitable for drilling mud production |
| | | | Dikwa (D4) | |
| 2013 | Oyegoke, T. | Adamawa state | Sangere, Yola | Yola clay is a low grade which contains high calcium ion with which can be beneficiated to API specification by addition of polypac. |
| 2013 | Olugbenga *et al.* | Niger Delta | - | Calcium based montmorillonite |
| 2013 | Nwosu, *et al.* | Enugu state | Udi | Sodium based montmorillonite |
| 2014 | Imuentinyan & Adewole | Edo state | Ovia & Ikpoba | Clays from the two areas belong to the montmorillonic clay family and has slight swelling ability due to high calcium content and can be used for drilling mud production. |
| 2014 | Nmegbu, C.G.J. | Rivers state | Egbamini (Emolga), Afam street (Phalga) and Oboboru (Onelga) | Clays from the three areas studied proved to be poor potential drilling mud materials far below the API standards for drilling mud formulation |

# Discussion of Findings

## Experimental leanings of the researchers

Table 2 shows the research efforts on bentonitic clays by researchers from 1964 up until 2014 and the experiment they did on Nigerian clays as well as the specific locations of the country where the clay samples were obtained. From Table 2, it is observed that most of the researches were done in the Northern parts of the country namely: Adamawa, Gombe, Kaduna, Sokoto, Yobe, and Borno states. This lends credence to the fact that the Northern part of Nigeria is a rich bentonitic clay field. Table 2 also shows that 44% of the researchers carried out clay sample characterization using techniques such as X-ray diffraction (XRD) and X-ray fluorescence (XRF) techniques while approximately 41% of the researchers carried out rheological analysis and only about 11% of the researchers studied both the rheological properties and the mineralogy of the clays.

## Mineralogical studies of the Nigerian bentonitic clay by researchers

The value of a raw material and its optimum use are recognized only by means of a thorough characterization. One level of characterization comprises of qualitative and quantitative data on chemical and phase (mineral) composition and fabrics. The behaviour of the material is the highest dimension of characterization that characterizes changes of a material and its properties in time and/or under fluctuating conditions. For material characterization, there are numerous analytical techniques and testing procedures. Nevertheless, for specific use, only a selection of these is applied. The comprehensive characterization of composition comprises of major and minor elements and major and accessory minerals as well. Partitioning of elements in coexisting phases and spatial distribution of mineral phases is crucial for eventual treatment and processing (Kuhnel, online). Table 3 presents the mineralogical analysis of Nigerian bentonitic clays from various researchers from 1964 up until 2014. Based on Table 3, it is observed that most of the researchers focused on determining the values of the major elements or oxides that mainly make up the montmorillonite content in the Nigerian clay which include: $SiO_2$, $Al_2O_3$, CaO, $Na_2O$, $Fe_2O_3$, among others. Interestingly, the $Al_2O_3$ to $SiO_2$ ratio of the Nigerian bentonitic clays ranges between 0.23 and 2.48. Thus, this ratio range compares favourably with Wyoming bentonite's $Al_2O_3$ to $SiO_2$ ratio of 0.38. In addition, as observed from Table 3, the oxides of Barium, Cobalt, Chromium, Copper, Nickel, Strontium and Vanadium in the bentonitic clay were not determined by any researcher. This may be owing to the insignificant role of the oxides in the montmorillonite content of the bentonitic clay. Finally, Table 3 also shows that almost all the researchers on the Nigerian bentonitic clay determined the loss on ignition (LOI) of the bentonitic clay. Their results indicate that the Nigerian bentonitic clay LOI ranges from 1.6 to 20.19.

**Rheological studies on Nigerian bentonitic clay by researchers**

Rheology is the study of the flow of matter, primarily in a liquid state, but also as 'soft solids' or solids under conditions in which they respond with plastic flow (Wikipedia, 2014). It applies mainly to substances or fluids with a complex microstructure such as drilling muds. It is necessary to study the rheology of muds since they are fluids that are in constant circulation in and out of a well under varying conditions. The rheology if not optimal would not permit the drilling mud to discharge its functions effectively. The major rheological properties of a drilling mud include the yield point (YP), the plastic viscosity (PV), the apparent viscosity (AV), the gel strength, among others. Table 4 shows the efforts of researchers from 1989 up until 2014 on the rheological properties of the Nigerian bentonitic clay samples they worked on. It is noticed from the Table 4 that most of the researchers concentrated on determining the values of the Plastic viscosity (PV), the gel strength, the pH, density and fluid loss of the clays while focusing less on the yield point and apparent viscosity of the clays.

**Results of Findings from Researchers**

Table 5 presents a summary of the findings of all the studies available on the use of Nigerian bentonitic clays as drilling fluid material from 1964 to 2014. It is seen that all the researchers agree that the bentonitic clays they studied contain montmorillonite. On close examination, all the studies show that the clays are of the calcium based category. Therefore, the Nigerian bentonitic clays can be graded as Oil Company Material Association (OCMA) grade bentonite; thus requires a lot of beneficiation to make them meet the API bentonite standard. Furthermore, Table 5 indicates that the researchers were unanimous in their findings.

## Learnings from the Review

It is observed that during the pioneering study years on the Nigerian clays as possible drilling fluids, local materials were mostly used for upgrading the clays to API bentonite standard. Though this trend was short-lived, it later resurfaced in the mid-2000 when Okorie (2006) used various local additives to formulate mud with the local clays. During the awakening years, it is observed that most researchers used mainly $Na_2CO_3$ and other polymeric materials to enhance Nigerian clay properties for use as drilling mud. It was also noticed that about 44% of the researchers carried out only an elemental analysis of the clays and drew their conclusions based on that. Again, a little over 40% of the researchers tilted towards the rheological properties of the clay when used as mud while 11 % of the researchers carried out both the elemental and rheological analysis of the clays. In all of the works reviewed, only Joel and Nwokoye (2010) carried out an economic analysis of what may be expected if some local or foreign materials they used in beneficiating the clays were to be done on a large scale. All this is depicted in Figure 4. In addition, most of the research on beneficiation has been adopting the one factor at a time (OFAT) experiment which gave a satisfactory result as only one factor was changed. However, this approach cannot be used to study the effect of two-factors at a time. Additionally, it is observed from this review that most clays in Nigeria are Calcium based bentonites and can only be useful as drilling muds when beneficiated. Finally, the preponderance of clay studies was mostly done in the Northern part of the country while there was just one research recorded in the south western part of the country.

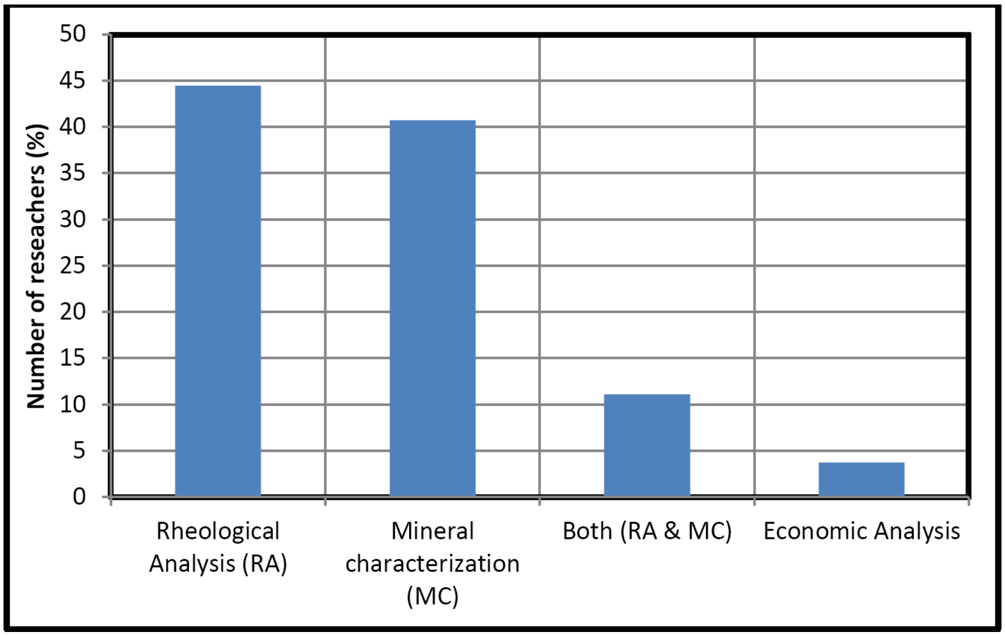


**Figure 4—Percentage leaning of various researchers on Nigerian clays for use as drilling muds**

## Suggested Research Directions for the Future

In view of the numerous areas x-rayed in this review work, the following are proposed as future research directions on this subject:

1. For a comprehensive approach, the mineral characterization and rheological studies on each clay deposit should be studied in one piece. This would provide a detailed view of the clay under consideration.
2. More researches should be done with regards to the economic analysis of beneficiating clays using local additives as against the imported bentonite clays. The cost of purchasing necessary additives for the treatment of the Nigerian clays should be compared with the cost of imported bentonite clay. Again, there should be an investigation concerning local bentonite mining cost, processing, packing, and transportation.
3. The viscoelastic properties of the beneficiated clays and their effectiveness in drilling wells domiciled in challenging downhole environments such as High Temperature/High Pressure (HTHP) wells, ultra deep waters, among others should be explored. The thermal stability of the beneficiated local clays is also an important area of concern.
4. The optimal amounts of each beneficiating additive required to standardize each local bentonitic

clay sample adaptable to the API requirements for bentonites should be studied; as well as the shelf life of the treated local clays.
5. The reserve estimates of the clays in each region would be another area of research concern since there are no unanimous estimates of the reserve.
6. Studies into the compatibility evaluation of local clays with different local beneficiating additives at different thermodynamic conditions are also a viable area of possible research.

## Conclusions

From the review, the following conclusions are drawn:

1. Nigeria has a huge reserve of untapped bentonitic clay deposits and this clay is predominantly Calcium based.
2. Most parts of the Nigerian state have had their clays studied in whole or in part for use as drilling muds. This led to the discovery that the addition of various additives in given proportions can upgrade Nigerian clays to the requirement of the API bentonite standard.
3. The present level of consumption of bentonite especially in wellbore drilling operations in Nigeria is high, most of which is imported leading to loss of huge revenues to the Nigerian economy.
4. Given the vast quantities of bentonitic clays in different areas of the country, more research in areas such as those suggested in the preceding section is welcome. By doing this, the subject would be brought to the front burner once again as a pressing issue.
5. The authors acknowledge that this work may not have captured all the works done in this area due to the fact that some of the works may not have been published or are still undergoing review for publication and are not available to the authors. However, the consolation is that this work gives an insight into what may be a rallying point for future works. Remember, the quest for perfection can be the enemy of intellectual progress.

### Nomenclature

*AAS* = Atomic Absorption Spectroscopy
*API* = American Petroleum Institute
*AV* = Apparent Viscosity
*AK-P* = Akanwu Powder (Locally sourced Trona)
*BDL* = Below Detection Limit
*BPHSP* = Burnt Palm Head Sponge Powder
*CMC* = Carboxyl Methyl Cellulose
*FTIR* = Fourier Transform Infrared Spectroscopy
*HTHP* = High Temperature High Pressure
*LOI* = Loss on Ignition
$Na_2CO_3$ = Sodium Trioxocarbonate (IV)
*OCMA* = Oil Company Materials Association (*Now defunct)*
*OFAT* = One Factor at a Time
*PAC* = Polyanionic Cellulose
*PV* = Plastic viscosity
*RMRDC* = Raw materials Research & Development Council
*SPDC* = Shell Petroleum Development Company
*XRD* = X-Ray Diffraction
*XRF* = X-Ray Fluorescence
*YP* = Yield point

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